PD interview Questions and Answers.

1. How to fix setup and hold violations?

a) setup fixes:

1. Upsize the cell
2. Swap the cells from HVT to LVT
3. Add buffer if net delay dominates the buffer delay
4. Pushing the Capture path
5. Pulling the launch path

b) Hold fixes:

1. Down size the cell
2. Swap the cells from LVT to HVT
3. Add buffer at D-pin of the flip flop
4. Pushing the launch path
5. Pull the capture path

3. How many corners you worked?

a) I worked on 10 corners.

4. On which corner you will fix setup?

a) For setup I read rc_worst corner.

5. On which corner you will fix hold?

a) For hold I read rc_best corner.

6. How to fix setup and hold violations if the base layers are fixed?

a) Setup:

1. Increase the metal width.
2. Route with highest metal layers.
3. Hold :
    1. Detour the net.
    2. Route with lower metal layers.

7.Detail explanation about SDC?

a) SDC (synopsys design constraint file) In SDC we have information like :

1. Clock definitions
   - create_clock -name {test_clk} -period 8.000 -waveform { 0.000 4.000 } [list [get_ports {scan_clk}]]
   - create_clock -name {my_clk} -period 4.000 -waveform { 0.000 2.000 } [list [get_ports {clk}]]
   - create_clock -name {my_clk_v0} -period 8.000 -waveform { 0.000 4.000 }
   - create_generated_clock -name div_clk -source [get_ports {clk}] -divide_by 2 [get_pins {clk_div_reg/Q}]
2. External input and output delays
   - Set_input_delay 0.5 –clock {clk1} – add_delay –max/min –rise/fall [get_ports { input ports or internal pins}]
   - Set_output_delay 0.7 –clock {clk1} –add_delay –max/min –rise/fall [get_ports {output ports or internal pins}]
   - Set_load –max 2 [get_ports { io ports}]
   - Set_drive –max 3 [get_ports { io ports }]
3. Max drv's
   - Set_max_transition 0.25 [get_ports { io}]
   - Set_max_capacitance 0.1 [get_ports { io}]
   - Set_max_fanout 40 [get_ports {io}]
4. Path exceptions
   - Set_min_delay
   - Set_max_delay
   - Set_multicycle_path 3 –from [get_clocks {launch clock}] –to [get_clocks { capture clock}] -setup

     Set_multicycle_path 2 –from [get_clocks {launch clock}] –to [get_clocks { capture clock}] -hold
   - Set_false_path –from [get_clocks {clk}] –to [get_clocks {clk_v}]
   - Set_clock_uncertainty 0.2 –from [get_clocks{clk}] –to [get_clocks {clk_v}]

8. Explain PD flow ?

a) Inputs to the PD are netlist(.v) , .sdc , .lef , .lib , captables, udf/cpf.

1. Performing sanity checks on input files. if all sanity checks are cleared then only we move further.
2. Floor planning :
   - define core area dimensions, calculate utilization and aspect ratio then
   - place IO ports then
   - place macros by following macro guidelines then
   - place physical cells
3. placement :
   - Global placement : std cells are placed based on hierarchy
   - Placement optimization: scan chain reordering, HFNS, congestion fixing , timing optimization, area optimization, power optimization.

- Detailed placement: In detailed placement all cells placed in a legalized manner.
- Early global route. To analyse congestion and timing after placement stage.

4. CTS (clock tree synthesis):

   In CTS clock will be built form clock port to all flip flop clock pins. And then clock tree optimizations will be done

5. Routing:
   - Global route: Tool try to finds shortest paths and assigns metal layers to it.
   - Track assignment: In track assignment each net will be assigned on track.
   - Detailed route: physical connections will be formed by using search and repair algorithm.
6. Sign off : PV , LEC , STA , IR.

9. What are the inputs for floor plan?

A) netlist (.v) , lef

10. After placement what will you check?

- congestion
- timing (drv's and setup)
- core area utilization
- Cells placed in a legalized manner or not.

11. What are the inputs for placement?

- netlist (.v)
- sdc
- lef
- def
- lib
- captables

12. After CTS what will you check?

- skew and insertion delay (SOURCE LATENCY)
- timing (drv's , setup and hold)
- congestion
- core area utilization
- cell legality

13. What are the inputs for CTS?

- Placement netlist (.v)
- Sdc

- Placement def
- Lef
- Lib
- Captables
- Spec

14. What are the inputs for routing?

- cts netlist (.v)
- cts def
- sdc
- lef
- lib
- captables

15. After routing what you will check?

- timing (drv's , setup and hold)
- drc's (design rule checks)
- core utilization
- cell legality
- congestion

16. what is skew ? and what are the different types of skew?

a) skew is the difference between the capture latency and launch latency. There are two types of skew. 1. Positive skew 2.negative skew.

Positive skew improves setup and degrades

hold, Negative skew improves hold and

degrades setup.

17. What challenges you faced in your project?

a) Congestion, timing and drc's.

18. How many iterations you did in floorplan?

a) I did upto 20 iterations.

19. What are Endcap cells? Why we use it?

a) Endcap cells are the physical only cells, which are placed at the boundary of the chip to avoid cell damage during fabrication and to provide nwell continuity to the std cells at the end of the rows.

20. what is the command for partial blockage?

a) CreatePlacementBlockage –type partial –density 65 –box {x1 y1 x2 y2}

21. why we use halo?

a) Halo is one type of blockage; it placed around the macros to avoid base drc violations.

22. who will give you channel space?

a) Manager or team lead

23. what is skew, useful skew and global skew?

a) Skew is the difference between capture latency and launch latency.

Useful skew is the technique that is improving setup without impacting the hold

violation. Global skew is difference between max latency and min latency.

24. what is cross talk?

a) cross talk is the transfer of a voltage transition from multiple switching net (aggressor) to another static or switching net (victim) through a capacitor.

25. what is EM?

a) Electron migration: The gradual displacement of metal atoms along with the charge carriers due to the flow of high density current is called electron migration.

It results in shorts and opens in the circuit. To overcome EM increase the width of the metal layers.

26. How many clocks you used?

a) There are 3 clocks in my design. One is main clock, test clock and generated clock.

27. How many power domains you have?

a) I did one project on Low power design. In that 3 power domains are there.
Two are switchable and one is always-on domain.

28. Tell me some DRC'S you fixed?

a) spacing issues , shorts and opens.

29. you worked on block level or chip level?

a) I worked on block level.

30. what block shapes you worked on?

a) I worked on three projects , in that two are rectangle and one is square.

31. you worked on 28nm and 14nm technology, what differences you found?

a) 1. No.of metal layers are increased from 28nm to 14nm.

2. DRC rules are increased.

3. Cross talk issues are more.

32. why we use .sdc file in placement?

a) After placement we will check setup violations, so while calculating timing from in-to-reg and reg- to-out paths it requires external input and output delays those are mentioned in sdc file. And also any path exceptions like multicycle path , false path and asynchronous paths. And clock definitions are required for calculating timing.

33. why we use RC values?

a) To calculate net delays we required RC values.

***34. In which nets you found EM violation?**

a) In data path nets and clock path nets also(if we didn't provide NDR'S).

***35. Where you have more EM? In DC or AC?**

a) In AC.

36. What version Innovus you used?

a) Innovus 18.1

37. Do you manually worked on CTS?

a) yes , I solved the setup and hold violations after CTS stage.

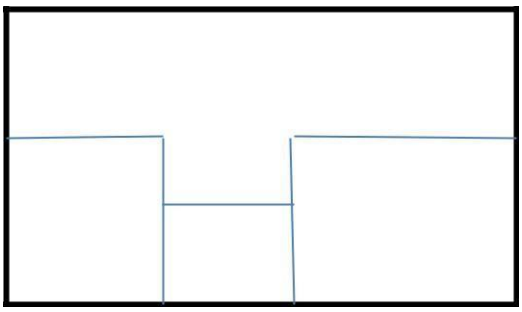
38. what you will do in floorplan?

a) Floor planning :

- define core area dimensions, calculate utilization and aspect ratio then
- place IO ports then
- place macros by following macro guidelines then
- place physical cells

39. Draw your block?

a)



42. In between macros std cells can we place?

a) At placement stage if any std cells are placed in macro channel , if we didn't get any congestion issues then we can place . if we are getting congestion in macro channel then put any hard blockage to avoid std cells placement.

43. what are the preplacement cells in your design?

a) In floorplan stage only Endcap cells and Welltap cells are placed.

44. How many clocks you have ? location of clocks? Frequency of design?

a) Three clocks are there in my design. IO ports are placed on top side of the block. Frequency is 500 MHz.

45. What you know about ECO?

a) After post route stage, the routed netlist is send to all sign off stages. At sign off stage if they found any timing , IR , lec and DRC's related issues, then sign off people wrote some ECO's (engineering change order) to solve those issues and send back to the PD people.

46. What is the relation between STA and PD?

a) In Physical design flow after Placement , CTS and Routing stage the STA people will involve to solve timing violations (DRV'S , setup and hold).

47. What you will do in CTS?

a) Before running CTS , we should provide some constraints like

- Specifying NDR's (Non default routing rules) 2w2s
- Specifying routing layers
- Specifying Skew and insertion delay targets
- Specifying clock buffers and clock inverters list
- Clock exceptions
- Clock grouping
- DRV targets

The main targets of CTS are to balancing skew and insertion delay. In CTS stage , clock will be built from clock port to all flip flop clock pins. And then clock tree optimization will be done.

48. what is the use of clock buffers and clock inverters?

a) clock buffers and clock inverters have equal rise and fall delays.

49. Which one you will prefer?

a) In my project I used both clock inverters and clock buffers. Clock inverter has good tran compare to clock buffers.

50. What do you meant by PBA?

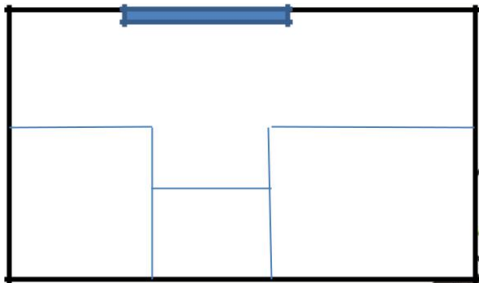
a) PBA (path base analysis) while calculating timing, tool will follow path base analysis . In path base analysis tool pick worst delay arc while calculating setup and min delay arc for the cells while calculating hold irrespective of the path.

51. If I give SOC level chip in primetime do you work?

a) Yes.

52. Draw your block? Where the pins are located? Which metal layers are used for your pins?

a)



pins located at top side of the block. Metal 4 is used for pins.

53. why you placed pins on top side , why not any other side?

a) The IO ports location is purely depends on data flow. Based on communication on top level chip , our block communication with other blocks will decides the port location.

54. How much is your latency?

a) 250 ps

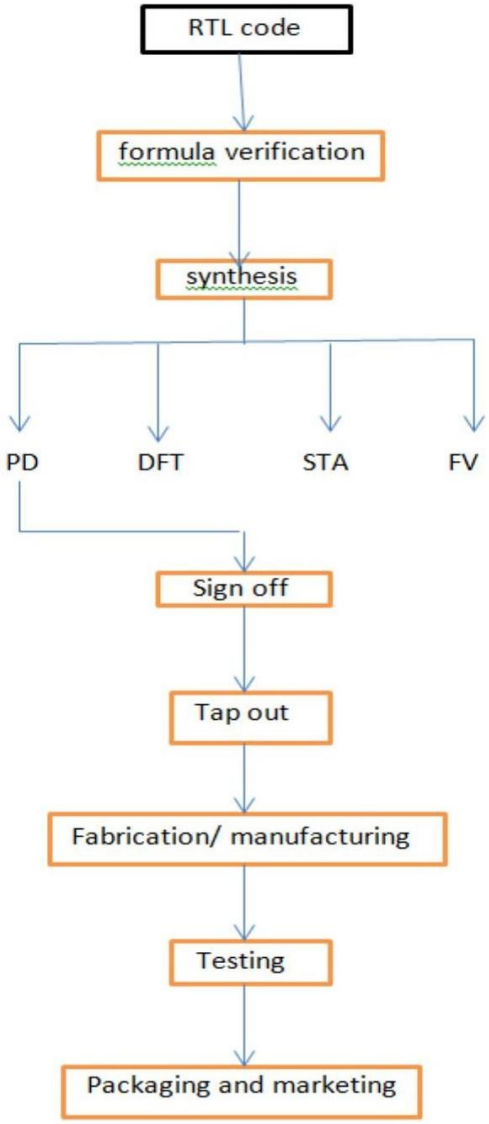
55. What are your responsibilities?

a) I involved from netlist to gds. and my role is to make good floor plan, solving congestion and solving timing violations.

57. What are the inputs of STA?

- netlist (.v)
- sdc
- lib
- spef

58. Tell me flow RTL onwards?



59. What is meant by empty modules?

a) for module, it has module definition and endmodule. It doesn't contain logic inside it.

60. Difference between HVT and LVT cells by transistor level?

a) LVT Cells has Low threshold voltage and high doping concentration, therefore less delay and more leakage power.

HVT Cells has High threshold voltage and less doping concentration, therefore more delay and less leakage power.

61. What consists of uncertainty where you're getting those values? Why we need to give those?

a) Pre CTS stage:

Uncertainty = skew + jitter + design

margin After CTS stage:

Uncertainty = jitter + design margin

We can get this value from top level people. To over constraint the design we consider uncertainty parameters.

64. What is your foundry ?

a) Samsung foundry.

67. What is useful skew ?

a) If the clock is skewed intentionally to resolve the violations, it is called as useful skew.

68. explain PD inputs ?

a) PD inputs are : netlist, sdc, lef, lib and captables, cpf/upf.

- Netlist : It contains the logical connectivity of the design. It contains Hierarchy information , module info , input and output pins info , wire info, instance info.
- SDC : It contains clock definitions , input and output delays , DRV'S and path exceptions.
- LEF : There are two types of lef's. 1. Technology lef 2. Cell lef.

  Technology lef contains physical information of all metal layers in the design. Cell lef contains physical information of the cells.
- LIB : It contains PVT information , power information (leakage and internal/short circuit power) , cell functionality and cell timing information.
- Captables : It contains resistance and capacitance values for the metal layers based on different width and spacing and different capacitors (coupling capacitance , fringe capacitance etc)
- CPF (Common power format): It contains power domain creation , operating conditions of the domains , low power cell (isolation , level shifter, switch cell, retension flop , always on buffer) rules.

69. What happens if lib and .v missing?

a) ERROR: (IMPIMEX-3): There is no verilog netlist found in init_verilog. The variable is either not specified or specified incorrectly. Check the location of Verilog file in the global file.

If netlist is missing tool won't load the design.

70. What happens if lef and lib missing?

a) It loads the netlist. But shows errors like

LIB missing: The design has been initialized in physical-only mode because the init_mmmc_file global variable was not defined. Timing analysis will not be possible within this session. You can only use commands that do not depend on timing data. If you need to use timing, you need to restart with an init_mmmc_file to define the timing setup, or you can save this design and use restoreDesign - mmmc_file <viewDef.tcl> to add the timing setup information.

LEF missing :

ERROR: (IMPEXT-3466):The technology layer information is not available. This can happen if either no LEF file is imported or an error occurs while reading the LEF file(s). The capacitance table file cannot be read without the layer information.

71. Explain sanity checks?

a) checkNetlist : The checkNetlist command will check the netlist , is it have any :

- Tri-state buffers
- Multi driven Nets
- Combinational loops
- No.of input floating's
- Empty modules

checkDesign –timingLibrary: This command is used to check the library. It shows errors like,

- Cells with missing timing information

checkDesign –physicalLibrary : This command is used to check the LEF file. It shows errors like,

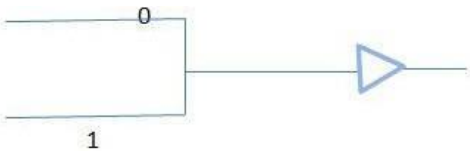
- cells with missing PG pin information , (it shows information related to missing physical information)

check_timing : This command is used to check the SDC.

- No clock or clock is ideal
- No input delay
- Unconstrained endpoints
- No drive assertion

timeDesign –preplace : This command is used to check the timing violations present in the design at synthesis stage only.

72. What happens if I have multi driven nets in my design?



a

Due to multi driven nets functionality will damage. Here we don't know that which value the buffer will drive. Multi driven nets should not be present in the design.

73. How will you place macros if you have 100 macros in your design?

a) By using design browser we can group the macros of same hierarchy, and then we will assign different colours to different hierarchies for easy identification. Then we will place particular hierarchy of macros at one place by following macro guidelines and based on communication by using flight line analysis.

74. Explain how power will supplied to macros and stripes?

a) The stripes will get power from power rings. And the macros will get power from stripes. Generally for power stripes we will use higher metal layers (M8, M9) but macro pins are on M3, M4, M5, M6 layers, so there should be stacked via from via3 to via9 to connect stripe to macro pin.

75. In my design I don't have cell density and pin density but I face congestion what could be the reason?

a) It may be due feed through paths present in our design.

76. Why setup violation occurs in placement stage?

a) In placement stage clock is ideal that means the skew is zero. Setup slack is difference of required path and arrival path. Setup slack = RT – AT. If RT value is less compared to AT value then we get the setup violation. In placement stage clock is ideal so less delay is present in clock path, so we get setup violations.

77. Why hold violation is not done at placement stage?

a) Because clock tree is not build at placement stage that means clock is ideal. Clock ideal means , skew is zero, so we can't get hold violations.

78. Explain macro guide lines?

a) 1. If you don't have reasonable rationale to place the macro inside the core area, then place macros around the chip periphery.

2. Communicating macros should be placed closure to each other.

3. If macro is communicating with IO ports then place macro near to IO ports.

4. Maintain proper channel spacing between macros.

5. Avoid criss cross connections.

6. Don’t rotate the macros but we can flip the macros.

79. Why we should not place macros at the centre of the core area?

a) Placing a macro centre of the core can invite serious consequence during routing due to a lot of detour routing, because macros are equal to a large obstacle for routing. Another advantage to placing the hard macros around the core periphery is it's easier to supply power to them, and reduces the change of IR drop problems to macros consuming high amounts of power.

80. What are all the placement commands?

place_design

optDesign –

preCTS

place_opt_desig

n

createPlacementBlockag

e checkPlace

refinePlace

report_utilization (checkFPlan –

reportUtil) reportCongestion –hotSpot

report_timing

81. Whether optimization will happen in CTS or post CTS?

a) Optimization will be done in post CTS stage.

82. Have you fixed skew? If so how?

a) I didn’t fixed skew. But I know how to fix skew i.e we should concentrate on each path while fixing skew because if you fix skew for one path then any other path gets violated. Fixing skew is nothing but adding buffers and removing buffers in the clock path.

83. What is ECO flow?

- load the routed data base
- remove filler cells
- source ECO file
- ecoPlace
- ecoRoute

- add filler cells
- generate spef
- send this data base to the sign off stages

for next time Eco's , we will load the previously updated routed database.

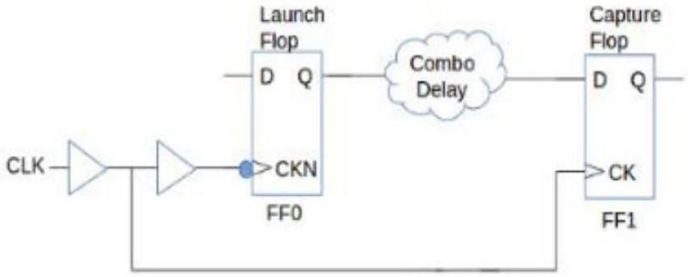
84. What are all sign off tools you have experience?

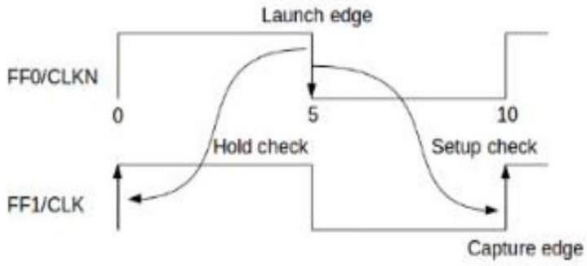
a) I have hands on experience on PD tool only. I don't have hands on experience on sign off tools.

85. what are half cycle paths? In half cycle path which one is critical? Setup or hold?

a) A path which requires only half cycle to capture the data. It is formed when data is launched on positive edge of the clock and captured on negative edge of the clock or when data gets launched on negative edge of the clock and gets captured on positive edge of the clock.

In such paths, setup check become more tight as setup gets only half cycle while hold constraint is relaxed by half cycle.





The falling edge occurs at 5ns and the rising edge occurs at 10ns. Thus, the data gets only a

half-cycle, which is 5ns, to propagate to the capture flip-flop. While the data path gets only half- cycle for setup check, an extra half-cycle is available for the hold timing check.

The hold check always occurs one cycle prior to the capture edge. Since the capture edge occurs at 10ns, the previous capture edge is at 0ns, and hence the hold gets checked at 0ns. This effectively adds a half-cycle margin for hold checking and thus results in a large positive slack on hold.

86. If you are facing congestion globally ,what is the reason for that?

a) The reason maybe bad floorplan.

87. After cts optimization how you will fix hold?

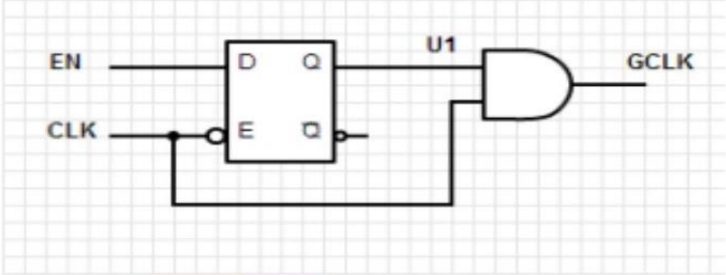
a) By skewing the clock path we will fix the timing violations. Hold can be fixed by pushing (adding buffers) launch path and pulling (removing buffers) the capture path.

88. what stage filler cells can be added?

a) After post route optimization we will add buffers in the design.

89. What is ICG? What is the advantage of ICG?

a) Clock gating is a very common technique to save power by stopping the clock to a module when the module is not operating. The advantage of ICG cell is , it reduces dynamic power consumption.



90. What you will fix first if there is setup , hold and DRV's?

a) First we have to fix DRV's and then setup and then hold.

91. In netlist we didn't find pg pin information then where will you find this ?

a) In def file we have information of special nets (VDD , VSS).

92. What issues you faced in placement ?

a) Congestion and timing violations

93. Did you get congestion between macro channel ?

a) No, I didn't get congestion between macro channel. Some standard cells are placed but congestion didn't occur. If I get congestion between macro channel then I will solve that congestion by applying hard blockage.

94. Which one is important skew or latency?

a) Latency is important than skew. If timing is met for both scenarios then which one you will choose?

1. skew =50ps, latency =200ps
2. skew =70ps , latency = 150ps

we will choose less latency i.e skew = 70 and latency = 150ps . because more latency , more buffers will be added due to this power consumption increases, area increases. So , to reduce power consumption and area we will give more importance to the latency.

95. What is the difference between soft , hard and partial blockages?

a) soft blockage : It won't allow std cells to place in that area during global placement. But it allows buffers and inverters during optimization.

Hard blockage : It won't allow std cells to place in that area during global placement and optimization also.

Partial blockage : It allow some percentage of std cells in that area.

96. How will you apply derates for setup and hold ?

a) For setup : early derates are applied to capture path and late derates are applied to launch path. For hold : early derates are applied to launch path and late derates are applied to capture path.

97. What is transition and what parameters affects it?

a) The time taken by the signal to switch from one state to another state is called transition time. Rise transition : The time taken by the signal to switch from 10% of its input to 90% of its output is called rise transition.

Fall transition : The time taken by the signal to switch from 90% of its input to 10% of its output is called fall transition.

The parameter effects transition are : High fanout , Long nets.

98. How to fix transition ?

a) 1. Add buffer if net delay dominates buffer delay

2. Upsize the driver

99. How to analyse timing reports ?

- First look at starting and ending points is both are triggered by the same clock or different clock. if it is different clock then check both clocks are synchronous or asynchronous.
- Then see the slack value . if slack is violating more than 1ns , then analyse why it is violating that much. This violation occurs maybe due to constraint missing like multi cycle path or half cycle path.
- If slack is less then see the report for which cell is taking more delay. And analyse why this cell taking this much delay i.e observe input tran and output cap of the cell.
- Find the no.of violating paths, if more paths violating with single start point and multiple endpoints then trace out diverging point from starting point. And do changes before diverging point.

- After analysing the reports use proper techniques to solve setup and hold violations.

100. What are NDR's ?

a) NDR's (non default routing rule) are used on clock nets . because clock nets are high switching activity nets so they effected more on cross talk and EM issues. In my design we used 2w2s . double width to avoid EM violation and double space to avoid cross talk violation

101. Why clock buffers are not used in datapath?

a) Because the size of clock buffer is large compared to normal buffer , it consumes more power and more area.

102. What is DRC and LVS?

a) DRC : DRC is design rule violations. In DRC it checks the design is meting DRC rules given by foundry or not.

LVS : LVS means Layout versus schematic . It is a method of verifying that the layout of the design is functionally equivalent to the schematic of the design.

103. Opens and shorts they are LVS issues or DRC issues?

a) opens and shorts are related to LVS issues.

104. DRC is passing and LVS is failing , does this scenario exists?

a) Yes

105. LVS is passing and DRC is failing ,does this scenario exists?

a) Yes.

106. What is cross talk noise ?

a) It is undesired change in the output values of victim due to switching in the input of aggressor. If one net is switching and other is at a constant value , the switching net may cause voltage spikes on other net. This is called as cross talk noise. Cross talk noise is evolving as a key source in degrading performance and reliability of high speed integrated circuits.

107. What is cross talk delay ?

a) When there is some delay in output transition of victim due to input transition of aggressor, it is called as cross talk delay. It occurs when some transition is happening in both the nets. Cross talk delay depends on the switching direction of the aggressor and victim nets too. If input transitions occur in same direction then output transition of victim becomes faster and if input transitions occur in opposite directions then output transition of victim becomes slower and delay is more which may violate setup time.

108. What are all the delay for given path consisting of two flops?

a) The delays in the data path are launch latency , Tcq (propagation delay of the flip flop), combinational delay

The delays in the clock path are capture latency, Tsetup (library setup time of a flop) , Thold (library hold time of flop) , uncertainty and CRPR.

109. what is start points and End points?

a) Start points are : Input port , Q-pin of the flipflop.

End points are : D-pin of the flip flop and Output port.

111. There is a two version of tools(v1 ,v2) while loading same data in v1 tool it won't see any congestion while load in v2 i m seeing congestion what may be the reason?

a)There are some variations in tool options from one version to another version. Some options will be enabled in one version by default , but in other it won't be enabled. So due this variations we can see the variations in reports.

112. Before routing your timing was good but after routing you find more violations what may be the reason?

a) 1. Before routing the net delays are estimated values, but after routing those are accurate. So we can see some violations from previous stage.

2. While doing routing we will enable some options like SI aware (cross talk),so to avoid cross talk issue some nets will be detoured due to this we get some timing violations.

3. During post route optimization some cells will be added in the design , due to this we may get congestion.

113. If there are setup violations in placement stage, then how will you fix ?

a) i. By applying path groups:

- group_path –name path1 –from starting points –to ending points
- setPathGroupOptions path1 -targetSlack 2
- optDesign –preCTS -incr

ii . Module constraint techniques :

- createRegion modulename or groupname 100 100 4900 4900
- createFence modulename 100 100 1234 1234

114. Due to what reasons we will see DRC's?

a) If track availability is less then all metal will route through that track then we get shorts, This may be due to more cell density at that area or pin density.

116. If you use normal buffers means what will happen?

a) For normal buffers the rise and fall times are not equal. When we use this normal buffers in clock path , min pulse width violations may occurs.

117. What settings you will do for CTS?

- create_route_type –name clkroute –non_default_rule 2w2s – bottom_preferred_layer Metal5 – to_preferred_layer metal6
- set_ccopt_property buffer_cells { CLKBUFX8 CLKBUFX12}
- set_ccopt_property inverter_cells { CLKINVX8 CLKINVX12}
- set_ccopt_property clock_gating_cells TLATNTSCA*
- create_ccopt_clock_tree_spec –file ccopt.spec

118. What is LEC?

a) LEC is logic equivalence check, LEC compares revised netlist and golden netlist , and it checks is both are logically equivalent or not that means functionality should be equal.

LEC inputs : Revised netlist , Golden netlist

and .lib LEC FLOW:

- read golden netlist and read revised netlist and then read the libraries
- specifying design constraints
- mapping
  resolving unmapped key points
- Compare process
  Debugging non-equivalent key points
- Report run statistics

119. What is antenna violation and ways to fix it?

a) When a long metal is connected to the gate terminal charge accumulation takes place on the surface of the metal during the etching of plasma. This charge tries to discharge at gate and damages the gate oxide.

Ways to fix congestion:

1. Metal jogging :

   This is preferred when the antenna effect occurs in lower metals. We split the metal near the gate and add a higher level metal parallel to that. Only higher level metal are preferred because during fabrication lower metals are fabricated first. If charge accumulation takes place ,it can be removed by process known as rinsing.

2. Antenna diode:

   This is preferred when the A.E occurs in higher metals. We connect it near the gate terminal in a reverse biased mode. As the charge flows breakdown occurs and no harm happens.

3. Break the net and add buffer on long nets.

120. What are Decap cells?

a) Decap cells are temporary capacitors added in the design between power and ground rails to counter functional failures due to dynamic IR drop happens at the active edge of the clock at which a high percentage of sequential and digital elements switch. At active edge of clock when the current requirement is high, these decaps discharge and provide boost to the power grid.

One drawback of decap cells is that they are very leaky, so the more decap cells the more leakage.

121. What is Ocv, derates, uncertainty, jitter and what makes it to occur?

a) OCV (On chip variations) occurs due variations in PVT (process, voltage and temparature). Ocv is considered for single PVT.

Derates : PD side estimation of OCV in percentage.

Uncertainty : uncertainty includes skew , jitter and design

margin. Jitter : deviation of clock edge from its ideal location.

122. What is uncertainty before and CTS and After CTS?

a) Pre CTS :

uncertainty = skew + jitter + design

margin post CTS :

for setup : uncertainty = jitter + design

margin for hold : uncertainty = design

margin

123. Suppose you have 2 clock transition value 40ps and 50ps ,and with both your timing is met then what value you will choose?

a) we choose 40ps . if transition is less power dissipation also less. And speed increases.

124. What is ndm and how u will generate the NDM, what inputs you need to generate NDM?

a) NDM (new data model

) how create NDM :

- create_workspace orca.dlib –use_technology_lib abc.ndm
- read_db eg.db
- read_db eg.lef
- check_workspace
- commit_workspace –output eg.ndm

Inputs to NDM :

- .db and .lef

125. What is min density and why it is required ?

a) If there are less no.of metals than specified in particular region then it results in min density violations. If there are min density violations it results in dishing effects it causes opens.

126. How to balance insertion delay?

a) i. upsizing the cells

ii . removing buffers

127. what is path grouping? How do we do net weightage?

a) path grouping can be applied for in2reg paths, reg2reg paths and reg2out paths. The tool will optimize these paths based on the weightage value given to it.

Command:

- group_path -name anyname(path1) -from start point list –to end point list – weight integer value(1,2 ,3 etc)
- refine_opt -path_group path1

  refine opt will do incremental optimization. In path grouping command we have different sub options. By using this path grouping technique we can optimize specific paths or nets.

128. what is scan clock and functional clock? How do we differentiate?

a) The frequency of test clock and functional clock is different. The functional clock has more frequency than the test clock. Because test clock is used in test mode to check the manufacturing defects of the flops and combinational elements(i.e flops and combinational elements are working properly or not), to test this not much frequency is required . the functional clock is used in functional mode to check the timing of the design.

By using set_case_analysis command we can differentiate which clock to propagate in our design.

129. Have you worked on multi clocks? How do we take care?

a) NO I didn’t worked on multi clock entering points. But I have idea if multiple clock entry points are there in design. We have two scenarios: multi clocks means clocks with different frequency . this clocks can have generated logic outside the block (i.e in top level design) or inside our block.

i) If clock is generated inside the block ,then the tool will automatically balance the insertion delay and skew for those clocks by considering information mentioned in the spec file (like through pin).

But if the divided logic is in outside the block (i.e if it present in top level) then we need to apply clock grouping technique to balance the skew between those to clocks.

Multi clocks with different source then those clocks called as asynchronous clocks. For asynchronous clocks no need to calculate timing.

130.we are getting clock net shorts, what may be possible reason?

a) may be at that area we have more no.of sequential elements. We are getting shorts means the available tracks are less than the required tracks this may be due to having more cell density at that particular area and main thing is most of the cells are flip flops. Congestion is not cleared at that area in placement stage.

131. what exactly happens in post CTS?

A) In post CTS , various optimizations will be done that includes: meeting DRV's , setup and hold, Area and Power optimization, congestion reduction.

132. How channel spacing between macros is kept based on what factors?

A) channel spacing between macro's = no.of pins x pitch / ((total no.of metal layers available)/2)

133.How do we reduce congestion?

a) congestion occurs mainly due to three reasons.

i. high cell density ii. high pin density iii. Bad

floorplan techniques for reducing congestion are:

i. cell density : if congestion is due to cell density ,then apply partial blockages at that area to reduce cell density in that region.

ii. Pin density : if congestion is due to pin density ,then apply cell padding techniques.

iii. bad floorplan (notches) : if we have congestion at macro edges , then apply hard blockages to reduce congestion.

Other techniques are:

1. specify a maximum density that controls how densely the tool can place cells in uncongested areas during wire-length-driven placement.

place.coarse.max_density (default is 0 and the tool spreads cells uniformly by default)

2. Specify a congestion-driven maximum utilization that controls how densely the tool can place cells in less congested areas that surround highly congested areas, so that the cells in the congested areas can be spread out to reduce the congestion.

place.coarse.congestion_driven_max_util (Default is 0.93)

When specifying the maximum density or congestion-driven maximum utilization,

choose a value between 1 and the overall utilization of the block. For example, if the

utilization of a block is 40 percent, you can choose values between 1 and 0.4.

3. Enable automatic density control, which derives optimal values for the maximum density and congestion-driven maximum utilization

place.coarse.auto_density_control (Default is true)

134. What is the procedure to generate ndm ?

a) icc2_lm_shell -f ndm_gen.tcl

135. what is difference between place.coarse.max_density and place.coarse.max_util?

A) max_density: specify a maximum density that controls how densely the tool can place cells in uncongested areas during wire-length-driven placement.

place.coarse.max_density (default is 0 and the tool spreads cells uniformly by default)

max_util: Specify a congestion-driven maximum utilization that controls how densely the tool can place cells in less congested areas that surround highly congested areas, so that the cells in the congested areas can be spread out to reduce the congestion.

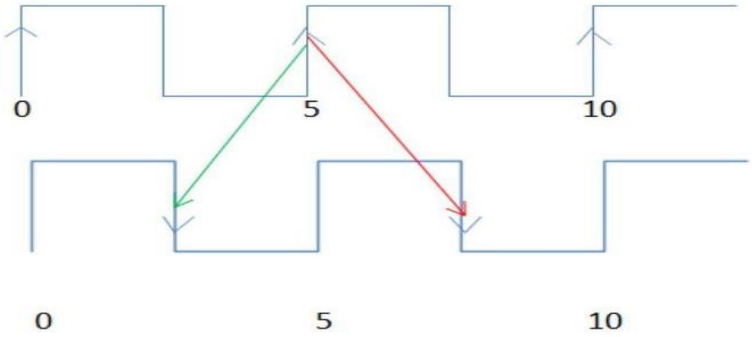
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utilization of a block is 40 percent, you can choose values between 1 and 0.4.

136. How do you check setup and hold for half cycle path?



A)

Setup check: 5ns to

7.5ns Hold check : 5ns

to 2.5ns

137. How do you fix setup and hold on same path i.e single start point and single end point? Reason?

a) There are two reasons for getting setup and hold on same path:

1. for setup we won't remove CRPR value because setup check is done at different edges, the impact of delay variation at one clock edge is different from delay variation on cell at different clock edge so we consider those variations(CRPR) for setup check.

Whereas for hold check is done at same edge, so we remove the CRPR value for

hold. Therefore due to this CRPR , we may get setup and hold violation on same

path.

2. if cell delay variations are more in data path from max lib to min lib , then we can get the setup and hold on same path.

Fixes:

1. we will swap the huge variation cells with less variation cells.
2. Reduce the common path by moving cells closure.

138. Multiple start points and single end point what is the reason for violation?

a) The reason for getting violation is that the data path is not optimized properly.

So in this case ,we trace the path from endpoint and find diverging point and try to fix the violation before diverging point by using techniques upsizing, swapping(HVT to LVT) ,adding

buffer if net delay dominates the buffer delay for solving setup. For solving hold use techniques like downsizing, swapping (LVT to HVT), add buffer at D-pin of the flipflop.

139. What is the difference between asynchronous clock and physically exclusive clock?

a) The clocks originate from two different sources are called as asynchronous clocks.

Physically exclusive clocks means, the clocks originate from same source but have only one entry point in the design.

140. Why do you apply MMMC from CTS when you already apply derates ?

a) derates are considered for intra chip variations i.e PVT variations on a chip (process
, voltage and temperature) , whereas MMMC (multi mode multi corner) is consider for environmental variations.

141. 3 flops – Flop 1 to flop 2 (-300ps slack) –flop2 and flop3 (+100ps slack) ? pre CTS how you will fix?

a)



In pre CTS stage , by upsizing or swapping the cells in the data path we can fix this

violations. Another method is using Time borrowing technique.

142. More buffers are added on longer nets and high fanout nets? What is the reason?

a) If we over constraint the max tran and max cap values, then over buffering will be happen.

143. What is useful skew?

A) Improve the setup slack without effecting the

hold. This can be done by pulling or pushing the

clock path.

144. Pulse width checks? What is input? On which cells we perform pulse width checks?

a) Minimum pulse width checks are done to ensure that width of the clock signal is wide

enough for the cell's internal operations to complete. i.e. to get a stable output you need to ensure that the clock signal at the clock pin of the flop is at least of a certain 'minimum' width.

If you need a formal definition of the term, it is the interval between the rising edge of the signal crossing 50% of VDD and the falling edge of the signal crossing 50% of VDD. If talking in terms of low signals, it is the interval between falling edge of the signal crossing 50% of VDD and the rising edge of signal crossing 50% of VDD.

Minimum pulse width depends on the technology node and the standard cell library design. You will have these modeled in your .lib file. Look for timing_type : min_pulse_width; in your liberty file. These will be defined for clock, reset and preset pins of a flop, or the enable pin of a latch.

SDC command:

- set_min_pulse_width –high 3.0 [get_clocks all_clocks]
- set_min_pulse_width –low 2.0 [get_clocks all_clocks]

145. If you don't mention in SDC? How will tool get to understand info?

a) we will have these modelled in .lib file. So the tool pick the data from this .lib file.

146. For antenna violation what extra take care we take when going from block level to top level?

a) when we have antenna violation in top level, then first observe port location of the blocks. We get this violation if any top level port is connected to block level port and the distance between these two are more, then one long net is connected between these two, due to this long net antenna violation occurs.

Fixes:

1. Insert antenna diode on the long net.
2. Metal jogging from lower metals to higher metals.
3. Insert buffer on long net.

147. what is CRPR ? Explain?

a) clock reconvergence pessimism (CRP) is a difference in delay along the common part of the launching and capturing clock paths.

By considering OCV(on chip variations) , the different derates are applied to launch and capture path of cells, the common path cells can't drive two different values at the same time, so to decrease pessimism we take the difference of the delays for common path cells.

148. What Vt cells you have used in design ? Did you meet timing in preroute stage?

a) Mainly SVT cells are allowed in the design at placement stage. Yes, meeting was met in preroute stage.

149. In CTS we have 50 cormers? Which corners you will pick for CTS? How will your run time be taken care?

a) In my project , at CTS stage I read rc worst corner for setup and rc best corner for hold. If we read more corners run time also more.

150. If your setup slack is worse in placement ? What could be the reason and what steps you will takecare?

a) There are many reasons:

1. The starting points and endpoints of flops are placed far apart , then due to long routing the path get violated. This happens because of scan chain reordering is not done in optimization stage.

2. During optimization if some nets are not optimized properly due to don't touch constraint.

Take care:

1. Do scan chain reordering.
2. Observe nets are optimized properly or not, if not then look on net property and change it to don't touch as false.

151. Once you setup and hold is fixed, Do you have to fix DRV – Tran violation? Will you fix or leave?

a) Yes , surely we have to fix DRV's. The delays taken for setup and hold are not accurate, the DRV's mask some hold violations. The setup and hold issues are met with inaccurate delays, so we get unexpected outputs. The chip behaviour is not as expected one.

152. Is fixing setup is difficult or hold? Which is difficult?

a) Setup fixing is more diffult, because setup is depends purely on design. For fixing setup violation ,we have to analyse the whole path why the violation occurs, what are the reasons for the violation. The analysis is difficult, it takes more time to analyse everything. The hold violation occurs due to skew and insertion(source) delay violations. The hold violation can be fixed by adding buffer at D-pin of the flipflop.

153. IR fixes ? How do you fix Dynamic and static IR drop?

A) Dynamic IR drop fixes:

1. If more no.of high switching activity cells are sitting at particular location, more IR drop will be there. So spread the cells in entire design.
2. Increase no.of stripes in the design.
3. Jog with higher metals.
4. Increase metal width.
5. Add via on missing via

locations.11111 Static IR drop fixes:

1. Swap LVT cells to SVT cells or HVT cells.

154. What is the difference between logically exclusive and physically exclusive clocks?

a) Physically Exclusive clocks: The two clocks originate from same source and have only one entry point in the design. By using set case analysis we can select which clock to propagate
in the design. At a time only one clock can exist in the design. These clocks won't affected by cross talk.

Logically Exclusive clocks: The two clocks originate from same source and have two different entry points in the design. By using MUX we can select the which clock to propagate in the design. These are affected by cross talk
.

155. Will scan chain reordering happen during CTS?

A) scan chain reordering is not happen in CTS stage. But it is not an hard constraint. Based on violation , if necessary we can do scan chain reordering in CTS stage also.

156. After your CTS is complete, you have more hold violations? What could be the reason?

a) After CTS we are getting violations means, the clock tree is not built properly. The skew is not balanced properly between flops. This will happen if skew target, insertion delay targets and DRV targets are not correct . if we miss any clock exceptions like float pin, through pin.

***157. If your congestion and timing is under control in placement , we have further scope for optimization? What is that exactly?**

a) The defined core area is more than the required.

158. OCV, AOCV and POCV in detail explanation?

a) OCV (On chip variation) : OCV occurs due to variations in PVT(process, voltage and temperature). OCV is considered for single PVT variations.

PVT variations are:

1. Process: During manufacturing all cells on the same wafer can't exhibit same behaviour due to PVT variations. All cells can't have same doming concentration level , gate width and oxidation .
2. Voltage: all cells on the same wafer can't get same voltage. The cells near the boundary can get more voltage than cells far from boundary , there maybe some IR drop in the metal layer due to long routing
3. Temperature: Theif high switching cells placed at particular region have more junction temperature than other cells at other regions.

***159. When your design is full of 32x driver cells? What issues we get?**

a) as drive strength increases, the size of the cell increases due to this ,It occupies more area and it consumes more power.

160. If you have setup 2k violation paths, hold 2k violation paths, What you will fix first and why?

a) First preference is given to setup because it depends purely on design, and difficult to fix. For fixing setup violation more analysis is required. Fixing hold is easy, we can fix the hold by adding buffer at D-pin of the flop if we have setup margin.

161. why can't we use AND gate instead of ICG cell?

a) If we use AND gate , glitches will be formed when enable signal toggles at positive level of the clock signal. So, to avoid glitches in the clock signal we will use ICG cell instead of AND gate.

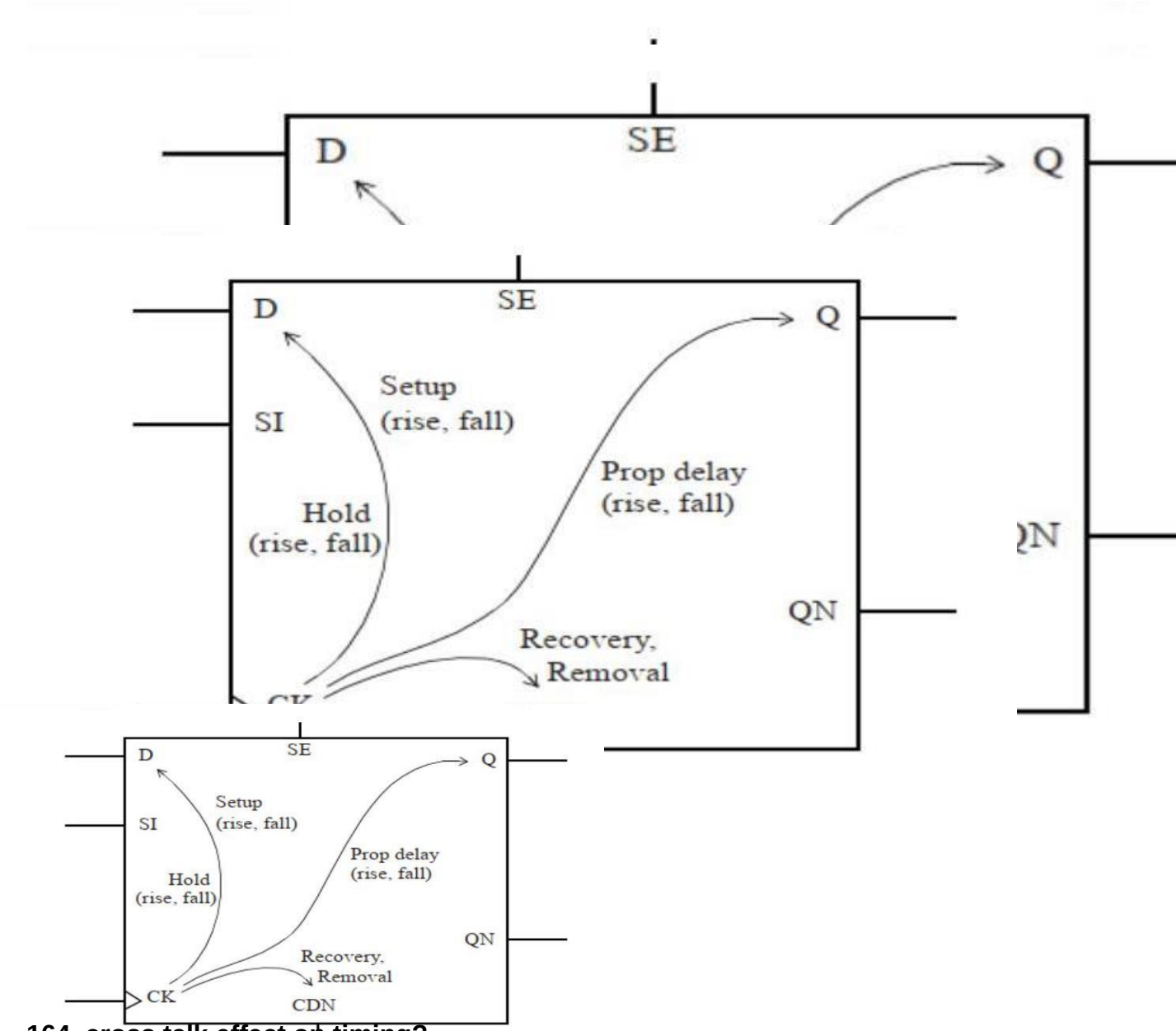
162. From Post CTS to PRO is there any timing degradation? If so why? If not why?

a) yes, timing is degrade from Post CTS to PRO(post route optimization) because actual signal routing is done in routing state. Before Routing stage virtual route is done, so timing is calculated by taking approximate delay for nets, but at PRO stage timing is calculated with actual net delays.

163. If we have timing violations how do we control or take care in CTS?

A) After CTS if we have timing violation means, the clock tree is not built properly. So first analyse the clock path i.e which drive strength cells are used and check all constraints are defined properly or not, if some constraints like clock exceptions (eg. Floating pin and through pin, ignore pin, exclude pin) , clock grouping (if our design has two or more main clocks) are missed then skew and insertion delay won't balanced properly, we get timing violations.

Take cares:



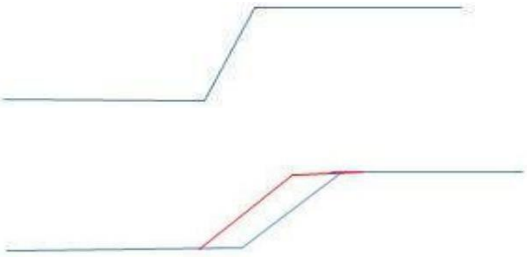
164. cross talk effect on timing?

A) Cross talk delay effects the setup and hold.

Scenario 1: If aggressor and victim net are switching in the same direction, then the delay of the victim net is reduced due to high switching activity of the aggressor net, it pulls the victim net so delay will be decreases. If this happens in data path it effects the hold violation. If this happen in clock path (in required path) then setup gets violated.

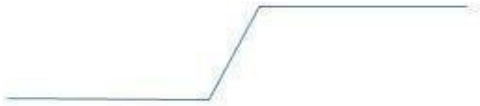
Aggressor: high switching activity net

Victim : less switching n

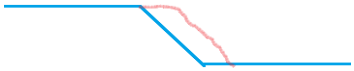


Scenario 2: If aggressor and victim net is switching in the opposite direction ,then the delay of the victim net is increases due to high switching activity of the aggressor net ,it push the victim net so delay will be increases. If this happens in data path it effects setup violation. If this happen in clock path (in required path) hold gets violated.

Aggressor:



Victim :



165. what is magnet placement?

a) To improve congestion for a complex floorplan or to improve timing for the design, you can use magnet placement to specify fixed objects as magnets and have ICC move their connected standard cells close to them.

You can specify fixed macro cells, a pin of a fixed macro cell, or an I/O pin as thFor the best results, perform magnet placement before the standard cells are placed.

To perform magnet placement, use the magnet_placement command with a specification of the magnets and options for any special functions you need to perform.e magnet object.

To instruct the magnet_placement cmd whether or not to pull cells toward the magnet objects, you can set the magnet_placement_fanout_limit variable to specify the fanout limit. If the fanout of a net exceeds the specified limit, the cmd does not pull cells of the net toward the magnet objects. The default setting is 1000.

Magnet placement allows cells to be overlapped by default. To prevent overlapping of cells, you can set the magnet_placement_disable_overlap variable to true, changing it from its default of false.